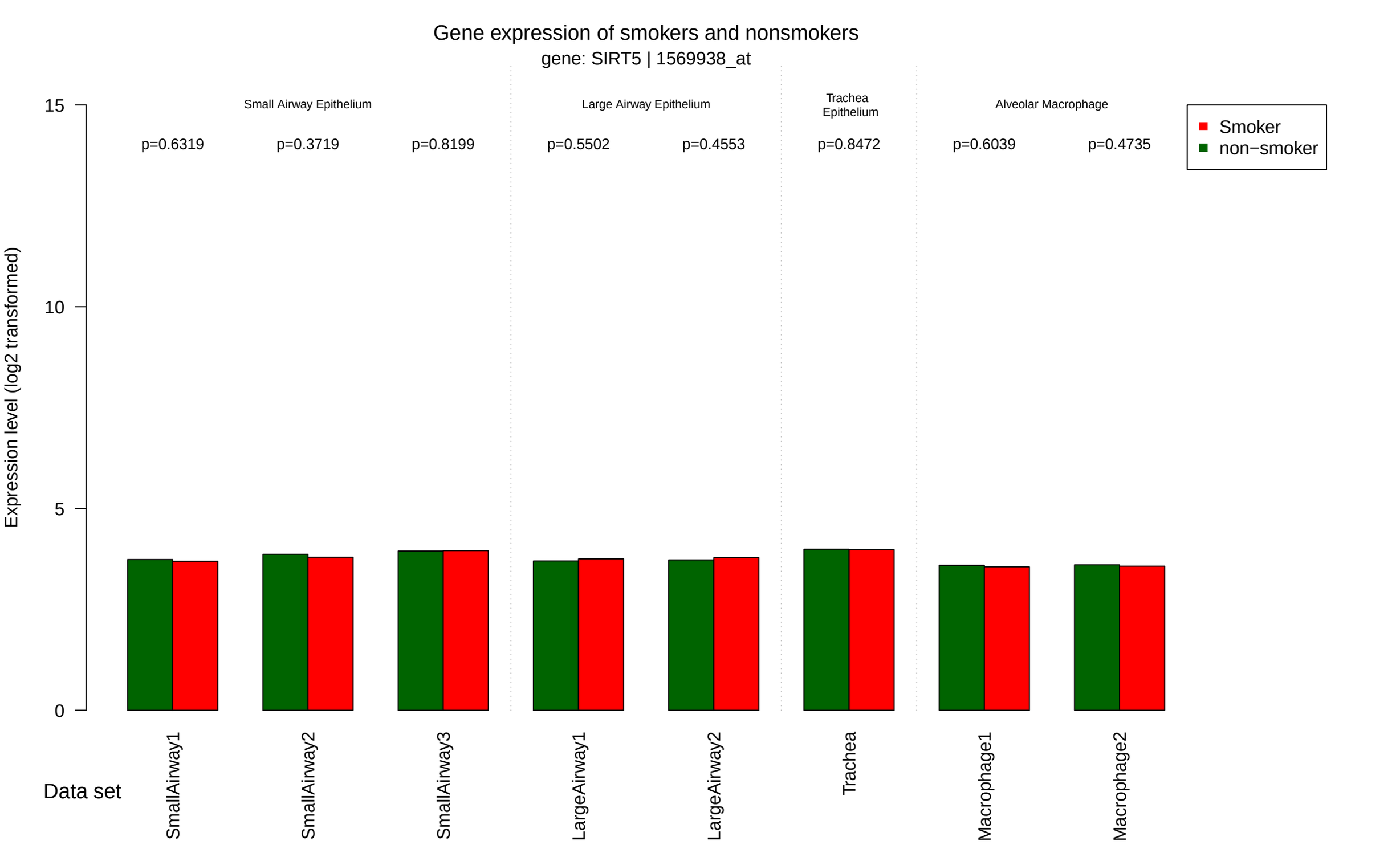
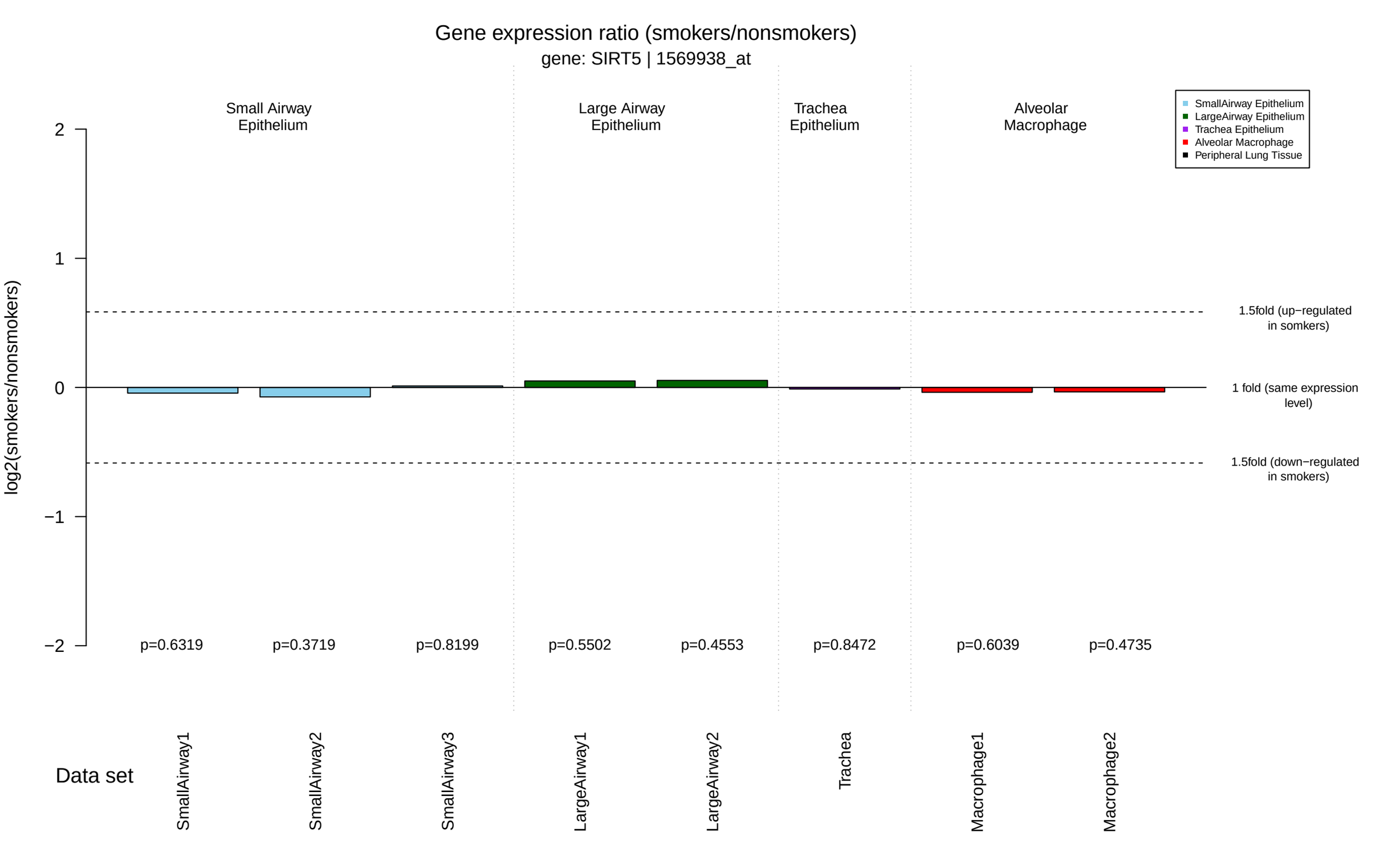
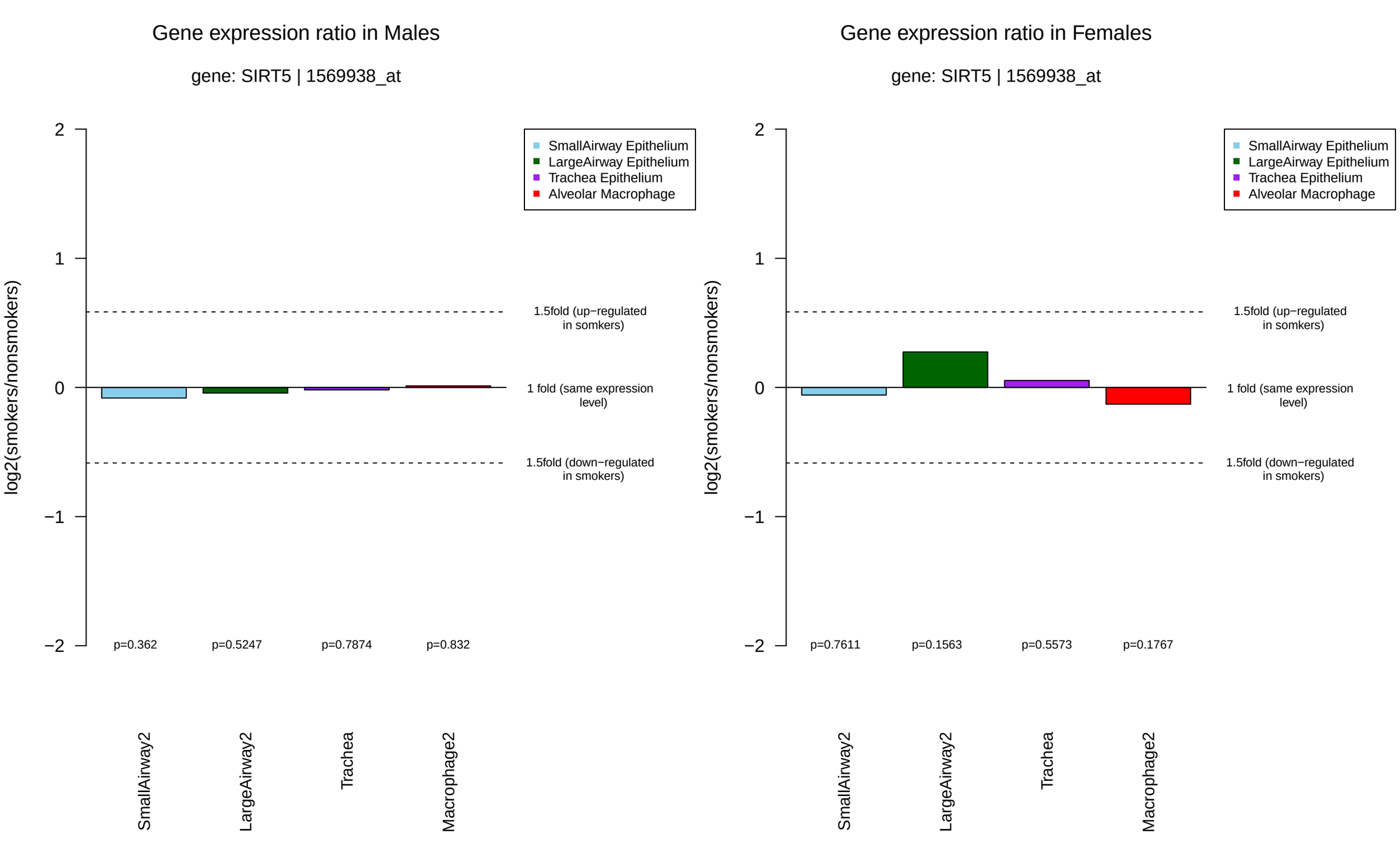
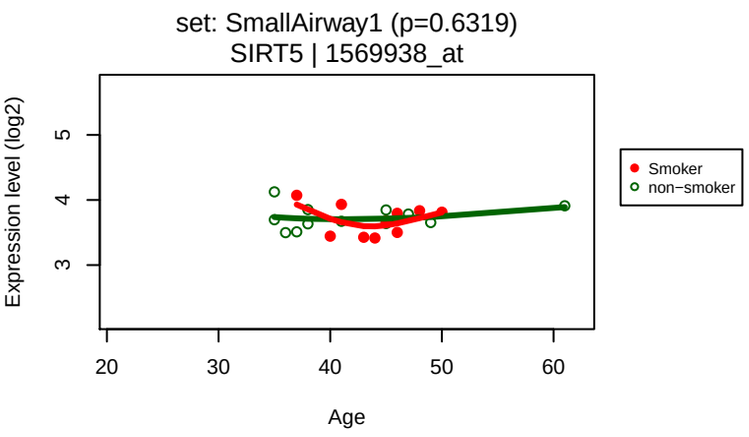
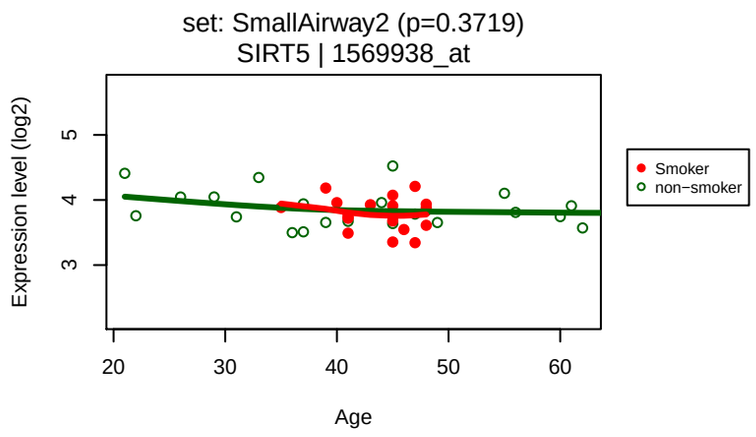
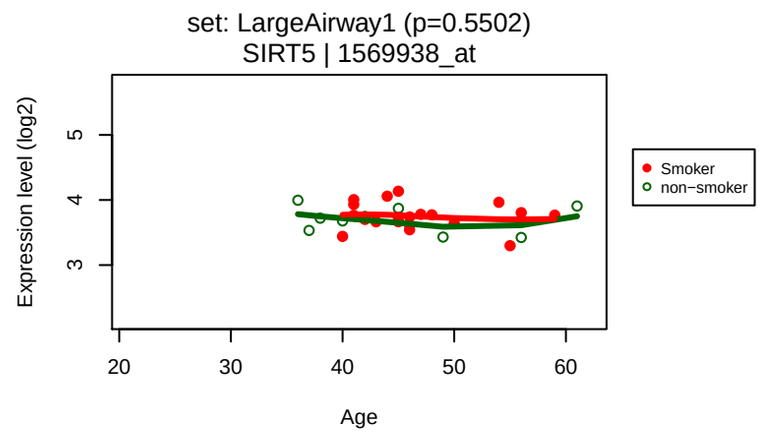

Gene expression of smokers and nonsmokers
gene: SIRT5 | 1569938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6319
p=0.3719
p=0.8199
p=0.5502
p=0.4553
p=0.8472
p=0.6039
p=0.4735
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

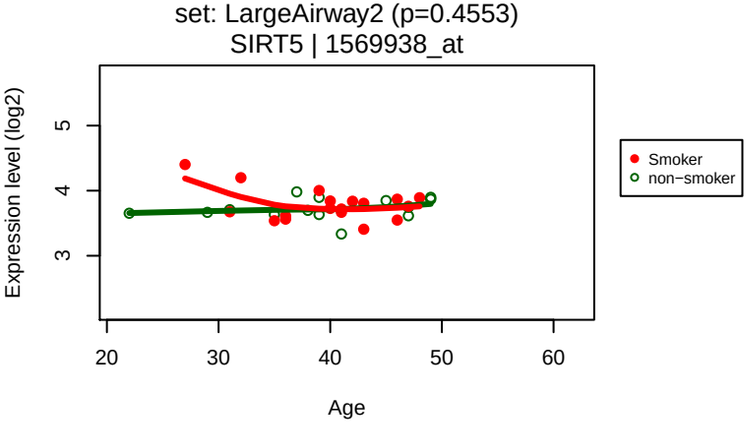

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 1569938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6319
p=0.3719
p=0.8199
p=0.5502
p=0.4553
p=0.8472
p=0.6039
p=0.4735
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

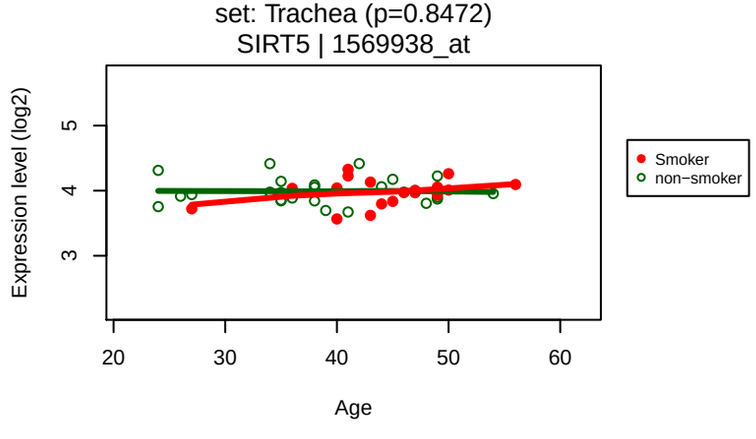

Gene expression ratio in Males
gene: SIRT5 | 1569938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.362
p=0.5247
p=0.7874
p=0.832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 1569938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7611
p=0.1563
p=0.5573
p=0.1767
SmallAirway2
LargeAirway2
Trachea
Macrophage2

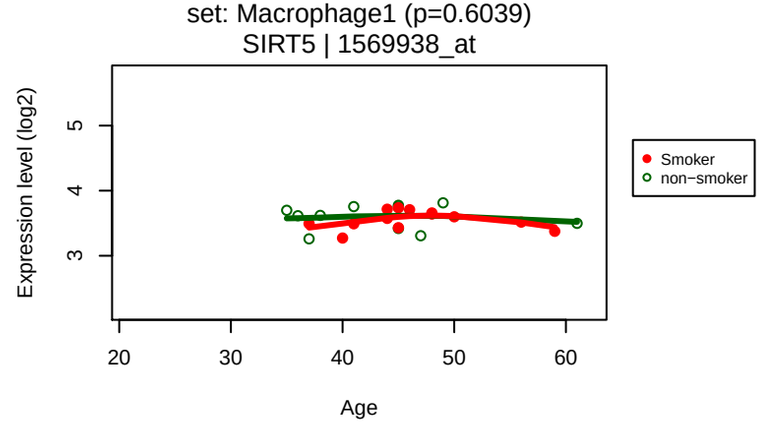

set: SmallAirway1 (p=0.6319)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker

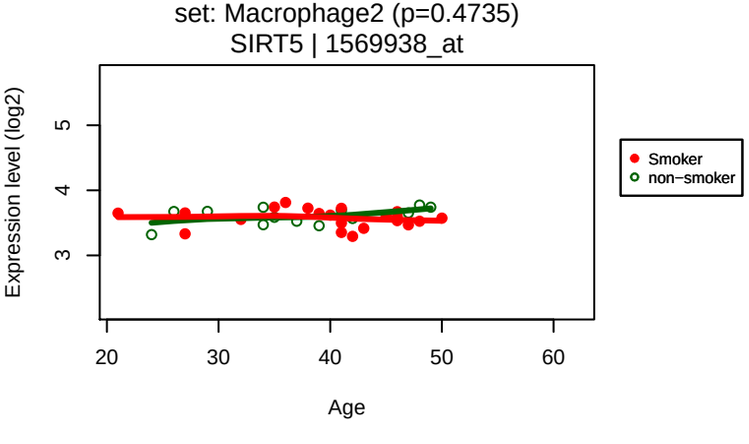

set: SmallAirway2 (p=0.3719)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker

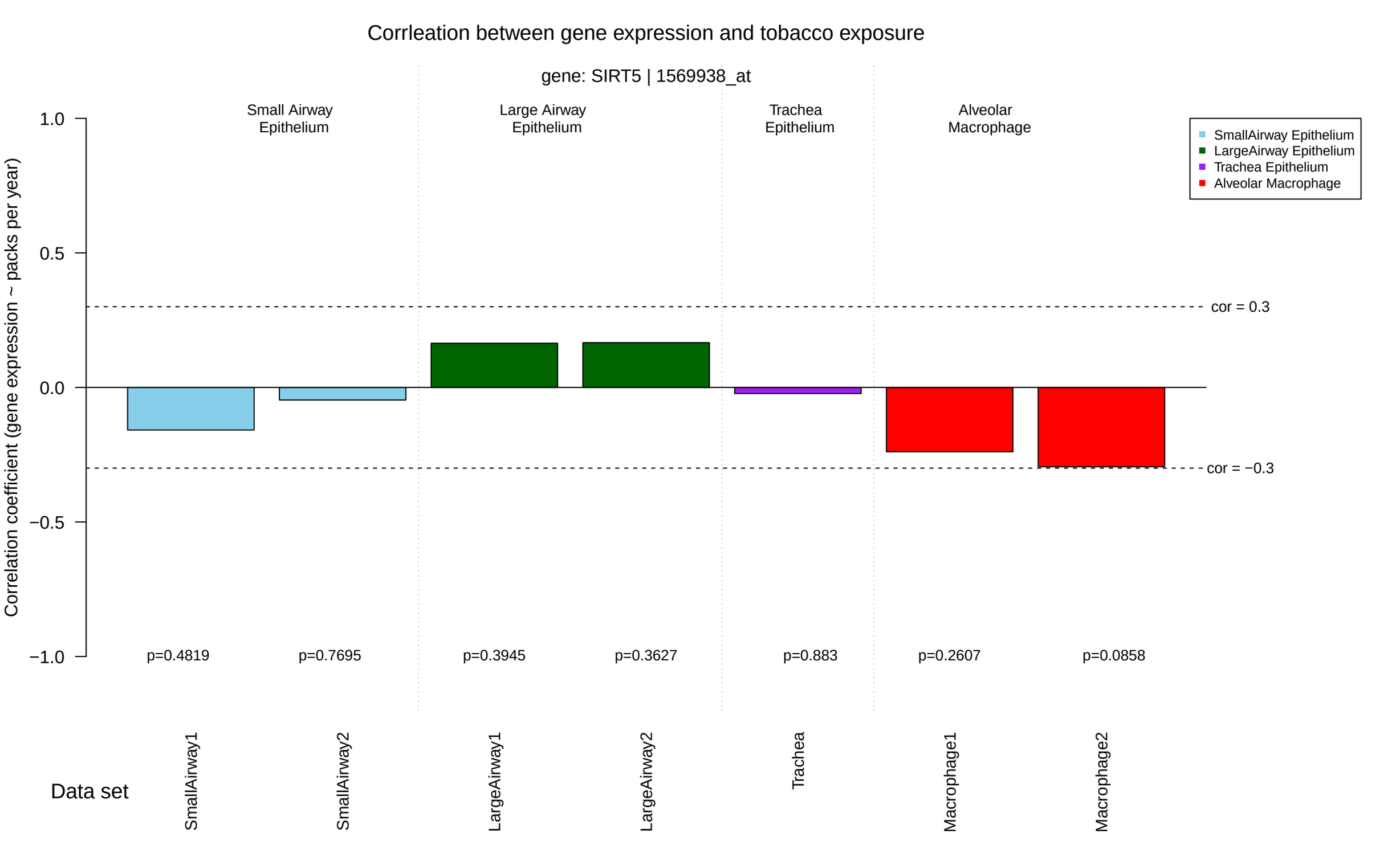

set: LargeAirway1 (p=0.5502)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4553)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8472)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6039)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4735)
SIRT5 | 1569938_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 1569938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4819
p=0.7695
p=0.3945
p=0.3627
p=0.883
p=0.2607
p=0.0858
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2